

Knowledge-Based Restaurant Recommender System

Project Summary

This system is a knowledge-based restaurant recommender that allows users to find restaurants based on explicit preferences such as cuisine, budget, and city. The application utilizes the Zomato Restaurants dataset and is developed using **Streamlit**, providing an interactive and user-friendly interface. The recommender does not rely on historical user data, thus mitigating the cold start problem commonly faced by traditional recommendation systems. It simply processes user preferences to generate the most relevant restaurant suggestions.

Evaluation Approach

User Feedback Collection

To assess the effectiveness of the system, users were asked to provide feedback after receiving restaurant recommendations. The feedback form captured the following:

- **Satisfaction Level:** (Satisfied / Neutral / Unsatisfied)
- **Relevance Score:** (0–10)
- **Open-Ended Suggestions:** Additional thoughts or improvements.

The collected feedback is stored in a CSV file (feedback.csv) for further analysis to enhance future iterations of the system.

Results Summary (Example Data)

Metric		Result
Total Responses	12	
Satisfied Users	8	

Average Relevance Score	7.6 / 10
Commonly Used Features	Cuisine & Budget
Suggested Improvements	Add map view, support for multiple locations

Observations

- **Simplicity & Clarity:** Users appreciated the easy-to-use interface, which made the process of finding restaurants seamless.
- **Trustworthy Results:** The explanation feature (providing reasons behind the recommendations) helped users trust the results more.
- **Filtering Features:** The most commonly used filters were based on **cuisine** and **budget**, showcasing their importance to users.
- **Requests for Improvement:**
 - A **map view** to display nearby restaurants.
 - The addition of more **visual content**, such as images or restaurant icons, to make the results more engaging.

Proposed Improvements

Area	Improvement
UI	Integrate a map view (using Folium or Streamlit's built-in map) for better visual representation of restaurant locations.
Dataset	Add support for restaurant coordinates to improve location-based filtering and the integration of a map.
Feedback	Implement a feature to highlight most liked restaurants based on feedback and ratings.
Ranking Logic	Allow users to choose between sorting restaurants by "popular" or "best-rated."

App Deployment

The application is hosted on **Streamlit Cloud**, making it accessible and easy to use for anyone with a browser.

Live Link: [Restaurant Recommender · Streamlit](#)

Files Included

- **app.py** – Streamlit interface file containing the app's interactive UI and logic.
- **scripts/** – Folder containing preprocessing and recommendation logic.
- **data/** – Folder containing both processed and raw datasets for restaurant recommendations.
- **feedback.csv** – File where collected user feedback is stored.
- **requirements.txt** – Python dependencies required for the app to run.
- **README.md** – Project summary and instructions for use.

Conclusion

The **Knowledge-Based Restaurant Recommender System** successfully meets the functional goals set in the project brief. It provides a user-friendly, interactive way to recommend restaurants based on specific preferences. User feedback indicates strong usability and satisfaction with the system, with constructive suggestions for future improvements. The integration of additional features such as a map view and image content will likely enhance the user experience further.